

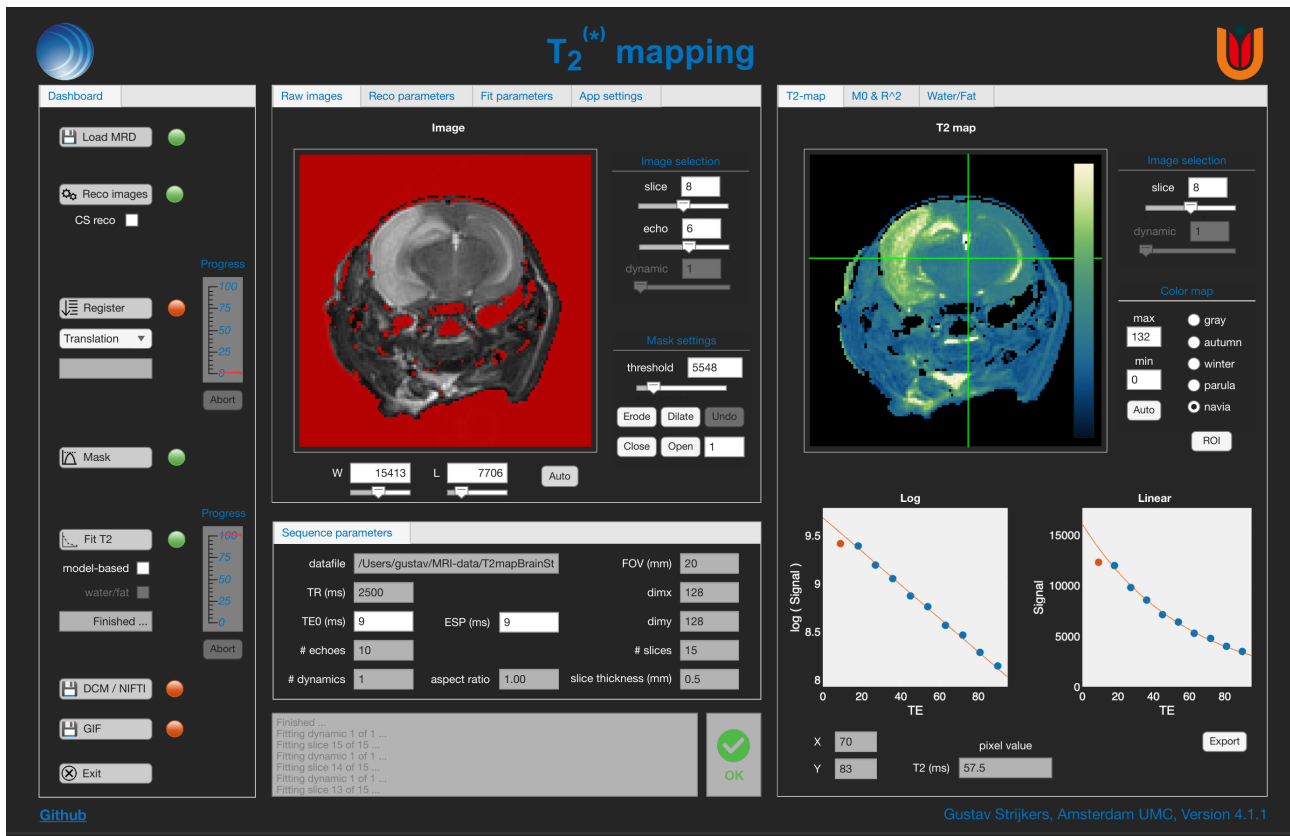


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1. Overview

T2mapp is a software utility and graphical user interface to reconstruct T2 maps from multi-echo spin-echo and T2* maps from multi-gradient-echo MRI acquired with an MR Solutions preclinical MRI system. It also separates water and fat in multi-gradient-echo data, and exports the maps as Dicom and NIfTI files, as a BIDS quantitative map, and as animated GIFs.

2. Contact

T2mapp and this manual are written by:

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3. Disclaimer

The software has been extensively tested using mouse and rat data acquired with an MR Solutions preclinical 7.0T/24cm system.

However, this does not warrant the functions contained in the program will meet your requirements or that the operation of the program will be uninterrupted or error-free.

In case of questions or issues, please contact Gustav Strijkers.



4. Installation notes

Software download

The Matlab app and a Windows standalone version (using the free Matlab runtime engine) can be downloaded from GitHub:

<https://github.com/Moby1971?tab=repositories>

Installation of the Windows standalone version

T2mapp_app_installer.exe

Will install the Matlab runtime engine and the T2mapp program.

Bart toolbox download

Compressed sensing reconstruction and model-based fitting in T2mapp require the Bart reconstruction toolbox, version 1.0.00 or newer, which can be downloaded from:

<https://mrirecon.codeberg.page/>

Without Bart, or with an older version, both are switched off; the message box says so for the model-based fit. With a GPU build of Bart, T2mapp runs them on the GPU.

Elastix registration

Image registration uses elastix, which comes with T2mapp for macOS, Windows and Linux. The macOS version is an Intel program: on a Mac with Apple silicon it runs only when Rosetta 2 is installed. When elastix cannot be run, T2mapp registers the images with Matlab's imregister instead, and says so at start.



Bart toolbox installation in OSX with MacPorts

- (1) Install Xcode from the Mac App Store

```
$ xcode-select --install
```

- (2) Install MacPorts (<http://www.macports.org/>)

It is recommended to install a newer version of gcc from MacPorts

- (3) Install packages

```
$ sudo port install fftw-3-single
$ sudo port install gcc14
$ sudo port install libpng
$ sudo port install openblas
$ sudo port install flock
$ sudo port install gmake
$ sudo port install coreutils
$ sudo port install libblocksruntime
```

- (4) Make

```
$ export PATH=/opt/local/libexec/gnubin:$PATH
$ CC=gcc-mp-14 gmake
$ sudo gmake install
```

Bart toolbox installation in OSX with HomeBrew

- (1) Install Xcode from the Mac App Store

```
$ xcode-select --install
```

- (2) Install HomeBrew

```
$ /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/
Homebrew/install/HEAD/install.sh)"
```

- (3) Install packages

```
$ brew install libpng
$ brew install fftw
$ brew install openblas
$ brew install gmake
$ brew install llvm libomp
```

- (4) Make

```
$ CC=gcc MACPORTS=0 gmake
$ sudo gmake install
```



Bart toolbox installation in Windows 11

(1) Install the Windows subsystem for Linux

Start a windows powershell as administrator and run the following command:

```
Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Windows-Subsystem-Linux
```

A system restart will be needed.

For more information, see:

<https://pureinfotech.com/install-wsl-windows-11/>

(2) Download Ubuntu Linux

Open PowerShell

To see a list of available Linux distributions for download, enter:

```
wsl -l -o
```

Install a Linux distribution:

```
wsl --install -d <distribution name>
```

You will be asked to create a user account the first time you start a Linux command prompt.

(3) Upgrade the Linux distribution

```
$ sudo apt-get update
```

```
$ sudo apt-get dist-upgrade
```

(4) Install Bart prerequisites (gcc 12 or newer)

```
$ sudo apt-get install make gcc libfftw3-dev liblapacke-dev  
libpng-dev libopenblas-dev gfortran
```

(5) Download Bart

```
$ wget https://github.com/mrirecon/bart/archive/v1.0.00.tar.gz
```

Recommended: use WSL2 (see e.g. <https://pureinfotech.com/install-wsl-windows-11/> on how to upgrade).

(6) Build Bart

```
$ tar xvfz v1.0.00.tar.gz
```

```
$ cd bart-1.0.00
```

```
$ make
```

```
$ sudo make PREFIX=/usr/local install
```



5. Running the software

Running in Matlab 2025a

The T2mapp software can be started from its root directory from the command line.

```
>> T2mapp
```

Additional licenses may be required.

```
MATLAB  
Image Processing Toolbox  
Parallel Computing Toolbox  
Signal Processing Toolbox  
Statistics and Machine Learning Toolbox
```

Running the Windows standalone

The Windows standalone version can be run from the start menu or the desktop icon.

Settings and log files

T2mapp keeps its settings in a file, t2mappSettings.json, in a folder of its own for every version of the app:

```
macOS      ~/Library/Application Support/T2mapp/<version>  
Windows    %APPDATA%\T2mapp\<version>
```

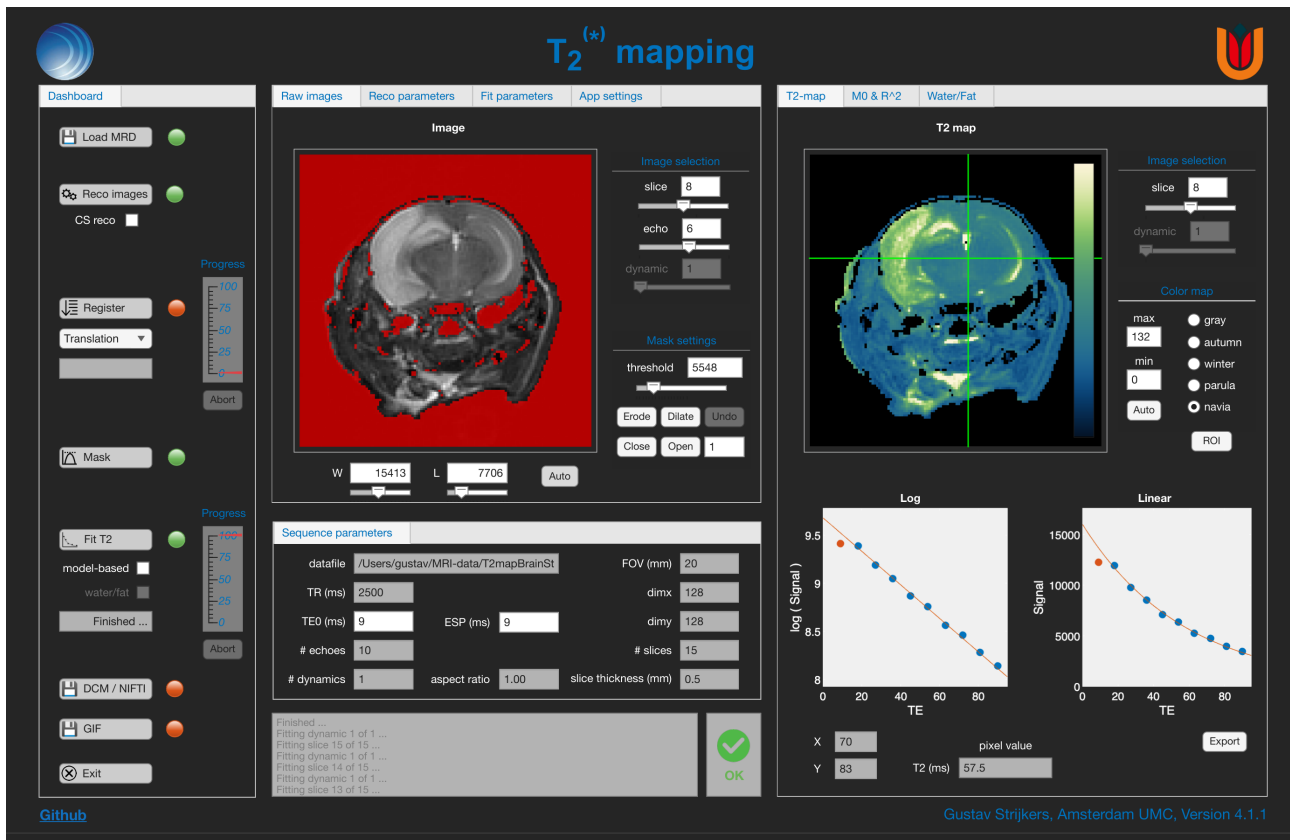
The file is written with the standard values the first time a version starts, and can then be edited by hand; changes take effect at the next start. It holds whether the app looks for a newer version and starts a parallel pool at start, the folders where loading and the exports start, the color scheme and the sounds, and the values the reconstruction, registration and fit fields and the color maps start with. A value that cannot be used is replaced by its standard value, and the message box says so.

Every session writes a log file, t2mapp-log-<date and time>.txt, to the same folder. The log keeps every message and the parameters of each reconstruction, registration and fit. Logs beyond the newest 50 (a number the settings file can change) are removed at start.



6. Basic operation

The T2mapp program operates from a single window with 5 panels.



Panel 1: Dashboard

This panel contains the task buttons and parameters that control the reconstruction process. The tasks run from top to bottom; registration and masking can be left out. A green light next to the task indicates that the task has been completed. When the light is yellow the task is busy. Red indicates not completed yet, or failed.

Panel 2: Raw images - Reco parameters - Fit parameters - App settings

This panel has 4 tabs: raw images, reco parameters, fit parameters, and app settings, where the sounds are switched on or off and the colors chosen, for the session.

Panel 3: Sequence parameters

This panel displays the relevant acquisition parameters; TE0 and ESP can be edited.

Panel 4: T2-map - M0 & R² - Water/Fat

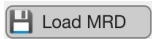
During and after fitting this panel shows the T2 maps. The M0 and R² goodness-of-fit maps are displayed in the second tab, and the third tab is for water/fat imaging.

Panel 5: Messages

Displays program messages and the status: OK, BUSY, WARNING or ERROR.



Step 1: Loading data

Press  to load the MRD raw data file.

T2mapp needs multi-echo spin-echo or multi-gradient-echo (FLASH) data, 2D multi-slice or 3D; 3D scans undersampled with a k-space table (pe1_order 5) can be read as well. Single-echo scans are refused, and the message box says why. MRD files written by P2ROUD or Retrospective can be loaded too.

For multi receiver coil data, select 1 of the MRD raw data files.

Relevant acquisition parameters will be shown in panel 3.

Sequence parameters					
datafile	/Users/gustav/MRI-data/T2mapBrainSt		FOV (mm)	20	
TR (ms)	2500		dimx	128	
TE0 (ms)	9	ESP (ms)	9	dimy	128
# echoes	10		# slices	15	
# dynamics	1	aspect ratio	1.00	slice thickness (mm)	0.5

The echo times follow from the first echo time (TE0) and the echo spacing (ESP) in panel 3. They are read from the scanner's own Dicom export of the scan when it is found, in .../Data/<study>/DICOM/<scan> beside .../MRD/<study>/<scan>. Otherwise ESP is taken equal to the TE for spin-echo data, or estimated from the sequence timing for gradient-echo data, whose TE0 then has to be entered before fitting. TE0 and ESP can be edited; maps fitted before a change have to be fitted again.

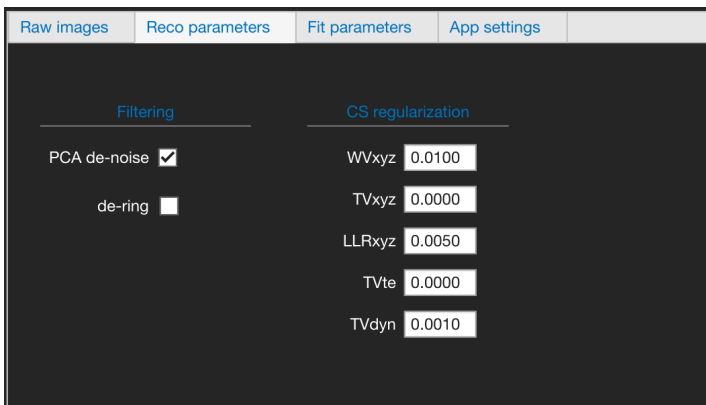
Step 2: Image reconstruction

Upon initial loading of the data, an automatic FFT image reconstruction will be performed.

For k-space undersampled data or to improve image quality by iterative regularization, a compressed sensing reconstruction of the data can be performed.

Select the CS reco check-box ☒ CS reco ☒ to enable compressed sensing reconstruction with Bart.

Compressed sensing reconstruction regularization parameters can be accessed via the Reco parameters tab.



Five compressed sensing reconstruction regularization parameters are available: wavelet in spatial domain (WVxyz), total variation in spatial domain (TVxyz), locally-low rank in spatial domain (LLRxyz), total variation for echo-time (TVte), and total variation in the dynamic domain (TVdyn).

Additionally a principal-component analysis denoising and a Gibbs ringing suppression can be engaged; PCA denoising is on by default, also for the reconstruction at load.

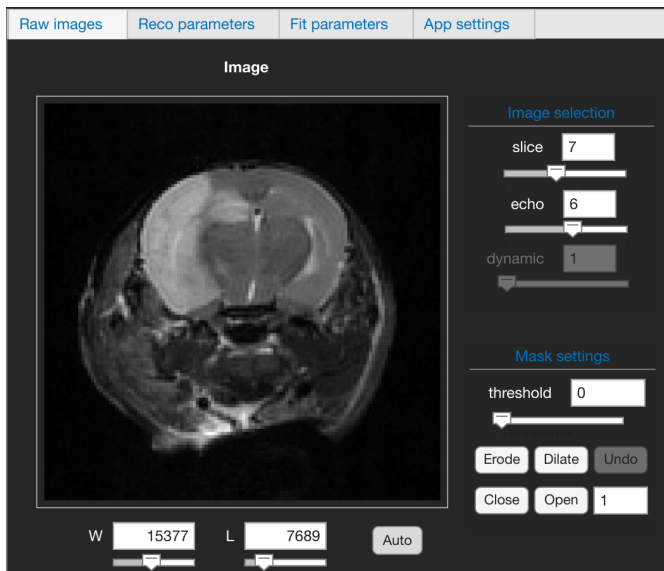
After setting the parameters, press to reconstruct the images.

PCA de-noise removes noise by principal component analysis, with the Marchenko–Pastur criterion of Cordero-Grande et al. (NeuroImage 2019), over the echoes and dynamics of each voxel, in a window chosen from the data. De-ring suppresses Gibbs ringing.

Step 3: Image registration

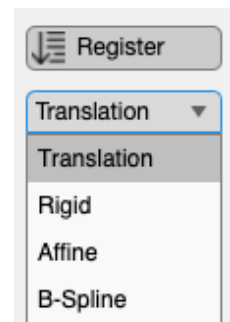
NOTE: this step usually can be omitted.

Use the sliders and/or edit-fields to inspect the slices and images as function of echo-time (TE) or dynamic/repetition. The image scale can be adjusted with the window (W) and Level (L) values and sliders; Auto sets them automatically. The mouse scroll wheel steps through the slices.



In case the images as function of TE are not perfectly registered, press  Register to perform an image registration.

There are 4 possible registration methods translation, rigid, affine, and b-spline, which can be selected via the drop-down menu below the register button.



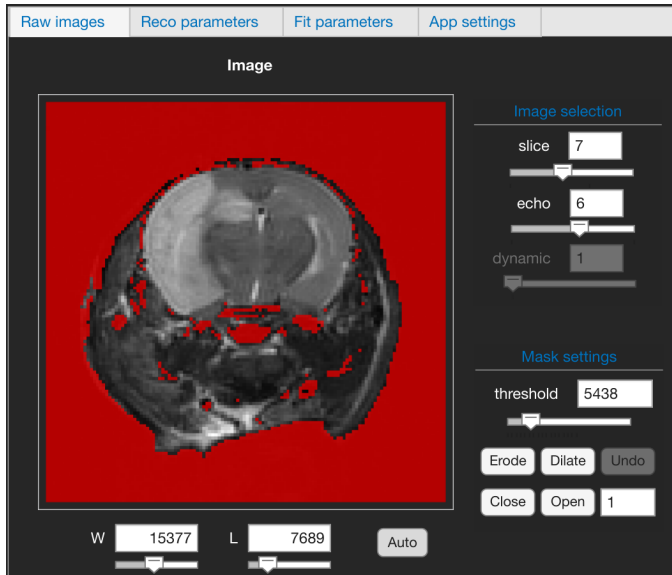
All echoes are registered to the first echo, with elastix, or with Matlab's imregister when elastix cannot be run (B-spline is then done as affine); the message box says which. The complex images are moved with them, so that a water/fat separation after a registration uses registered data. The progress and the time left are shown in the dashboard, and Abort stops the registration.

Step 4: Masking the background in the images

To prevent unnecessary fitting of the background pixels, mask the images first; without a mask every pixel is fitted.

Press  Mask to perform an automatic masking via thresholding, with a threshold for each slice (Otsu's method).

The result will look something like the figure below in which the red pixels indicate the regions that will be skipped during T2 fitting.



In case automatic thresholding is not satisfying, the threshold can be set with the slider or the threshold field; it then applies to all slices. The mask can be optimized by the morphological operations Erode, Dilate, Close and Open, on all slices at once, with the size in pixels set in the field beside Open (1 to 10). Undo takes back the last of these operations. They act on the red region: Erode shrinks it, so that more pixels are fitted, and Dilate grows it.



Step 5: Fitting T2 maps

Press the Fit parameters tab to inspect some fit settings.

Slice selection		Dynamics selection		Goodness of fit	
first slice	1	first dynamic	1	min R-square	0.80
last slice	15	last dynamic	1		

Echo selection							
☐ 1	☒ 9	☒ 17	☒ 25	☒ 33	☒ 41	☒ 49	☒ 57
☒ 2	☒ 10	☒ 18	☒ 26	☒ 34	☒ 42	☒ 50	☒ 58
☒ 3	☒ 11	☒ 19	☒ 27	☒ 35	☒ 43	☒ 51	☒ 59
☒ 4	☒ 12	☒ 20	☒ 28	☒ 36	☒ 44	☒ 52	☒ 60
☒ 5	☒ 13	☒ 21	☒ 29	☒ 37	☒ 45	☒ 53	☒ 61
☒ 6	☒ 14	☒ 22	☒ 30	☒ 38	☒ 46	☒ 54	☒ 62
☒ 7	☒ 15	☒ 23	☒ 31	☒ 39	☒ 47	☒ 55	☒ 63
☒ 8	☒ 16	☒ 24	☒ 32	☒ 40	☒ 48	☒ 56	☒ 64

The first and last slice that will be fitted can be selected.

The first and last dynamic that will be fitted can be selected.

Min R-square is the lowest goodness of fit (R^2) accepted: where a fit is worse, the T2 and M0 values of the pixel are set to zero.

The bottom part of this tab lists the echoes. Only the echoes ticked are fitted, which can be used to leave out, for example, the first echo (left out by default for spin-echo data), the last echoes below the noise level, or the even or odd echoes.

Press  to start the T2 fitting; Abort stops it.

Each pixel is fitted by nonlinear least squares (Levenberg–Marquardt) to the equation

$$\text{Signal} = M_0 \exp(-TE/T_2)$$

Model-based fitting

Tick the model-based checkbox  to perform a model-based reconstruction of spin-echo data with Bart.

It fits T2 and M0 to the k-space directly, so it gives no R^2 map and does not apply min R-square; 3D data are fitted volume by volume.

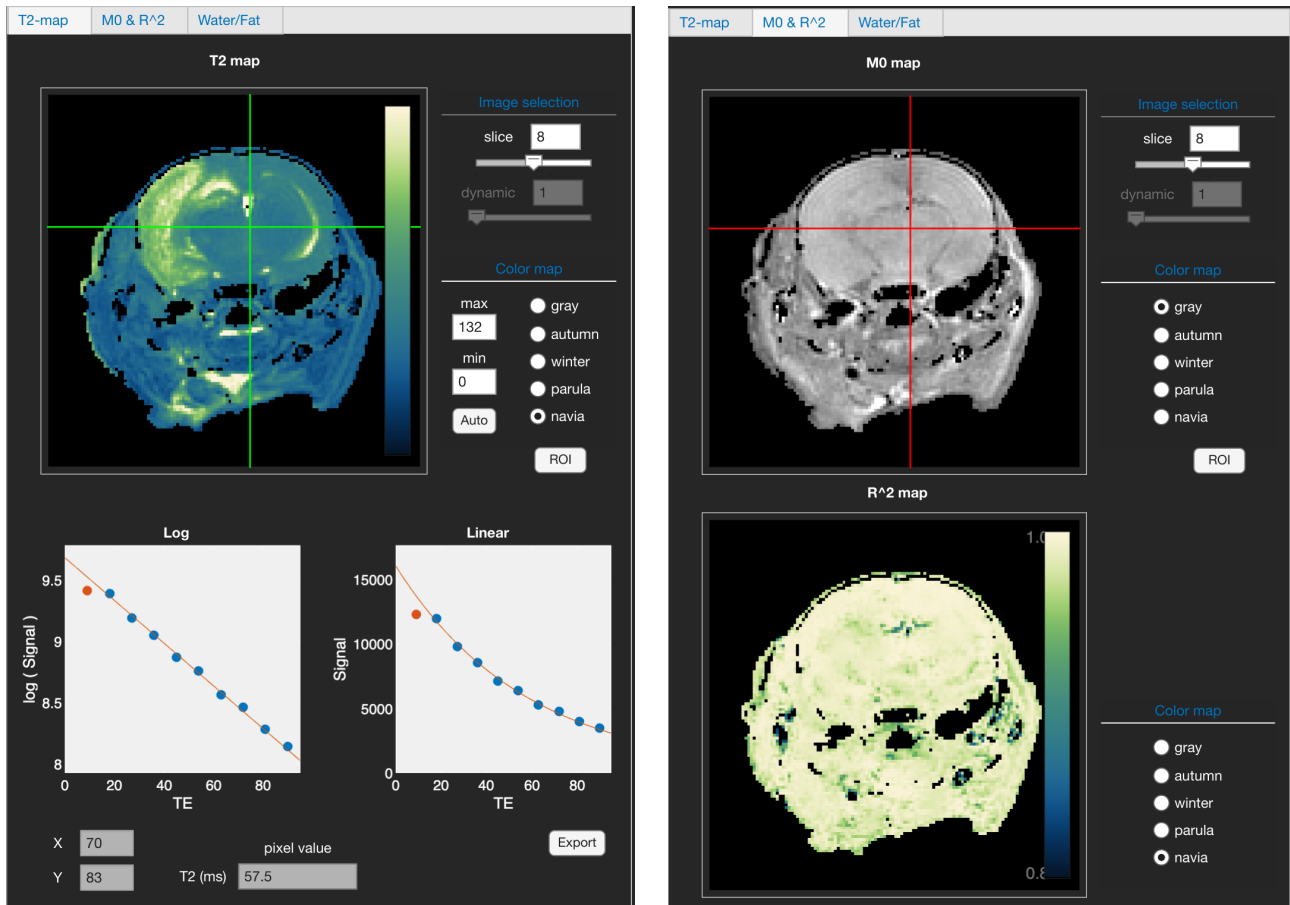
Water/fat separation

DIXON-based water/fat images can be reconstructed for multi-gradient echo data by checking the water/fat box . The water and fat maps, in percent, are in the Water/Fat tab.



Step 6: Fit results

The T2 maps will be shown in the T2-map tab. An M0 map and R² map are shown in the second tab. A color map can be chosen for each map.



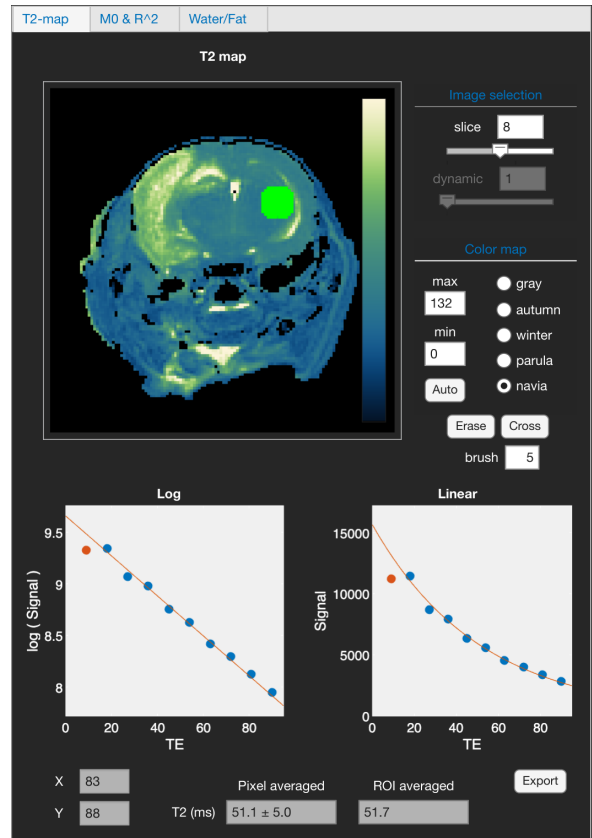
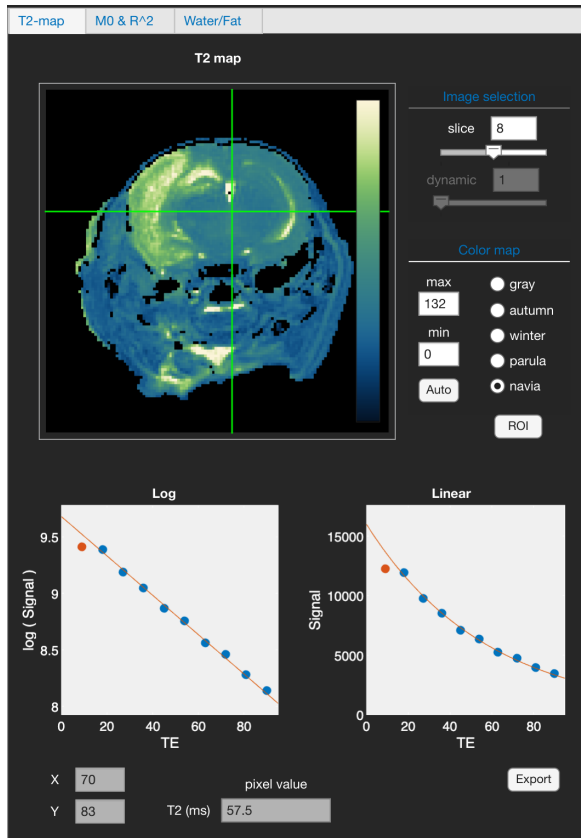
The T2 map is shown from min to max; Auto sets max to twice the mean T2 of the slice shown, and min to 0. The M0 map is scaled to the 99.5th percentile of the whole map, so that a few bright pixels do not darken it. The mouse scroll wheel steps through the slices of the maps. For gradient-echo data the T2 map is a T2* map.



Step 7: Extracting T2 and T2* values from pixels and ROIs

When you click on the T2 map with the mouse cursor a cross-hair will be shown. The $\log(\text{Signal})$ and Signal as function of TE (symbols) and the fit result (red lines) of the cross-hair center pixel, as well as the fitted T2(*) value of that pixel will be shown. A click outside the mask shows nothing.

A red symbol indicates that this TE was omitted in the fitting process, according to the list in the Fit parameters tab (see step 5).



You can toggle between the cross-hair and region-of-interest (ROI) values by pressing the button. A single ROI can be 'painted' on the T2 map in one or more slices by mouse clicking on the image, with the brush size set in the brush field.

For the ROI, two T2(*) values are calculated:

- Pixel averaged: the harmonic mean of the T2(*) of the fitted pixels, with its spread:

$$T2_{ROI} = 1 / \text{Mean}(R2_{\text{pixel}})$$
- ROI averaged: The T2(*) equation is fitted to the average signal intensities in the ROI, producing a single T2(*) value for that ROI.

The ROI can be removed with .



Step 8: Exporting the maps

There are two ways to export the maps, and one for the ROI data.

(1) DCM / NIFTI



Exports the T2, M0 and R² maps, and the water and fat maps when they were made, as Dicom series for further processing in 3rd party software. When the scanner's own Dicom images of the scan are found, their headers are used and the maps are written beside them, in the DICOM folder of the study. Otherwise the headers are made from the MRD data, with the orientation and position from the scan angles, in a folder that is asked for.

To prevent discretization into integer values in the Dicom files, the T2, R², water and fat maps are multiplied by 100, or by less where a map would not fit. The factor is written into the Dicom header (Rescale Slope), so that viewers show the true values.

The maps are also written as NiftI files, in the Nifti folder beside the Dicom folder, and the T2 map as a BIDS quantitative map, in BIDS/derivatives/t2mapp: T2map, or T2starmap for gradient-echo data, in seconds as BIDS defines it.

(2) GIF

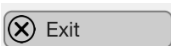


Exports the maps as animated GIFs, one frame per slice, or one GIF per slice with a frame per dynamic when there are several dynamics, in the color maps and scales shown, together with pictures of the app window, into the GIF folder of the study, or into a folder that is asked for.

(3) ROI data

The Export button of the T2-map tab writes the echo times and the mean signal of the cross-hair pixel or the ROI to a CSV file, roiData-<scan>.csv, in a folder that is asked for.

Step 9: Exit

Press  to shut down the program.

Closing the window does the same, also while the program is busy.